

R Practical 1: inspect, summarise and visualise

Data Science for Health Systems | Washa Takwimu 2026 Short Course | Day 1

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From a decision problem to a first piece of evidence

Day 1 learning outcome

“Produce a reproducible first summary and graph in R and state one limitation.”

By the end of this session you will have:

- Inspected three linked health data files
- Summarised one indicator, with uncertainty
- Made a fully labelled `ggplot2` graph
- Written down one limitation

Day 1 Decision problems + first summary

Day 2 Data quality, joining, auditing

Day 3 Maps, access and equity

Day 4 Measures, design, performance

Day 5 Decision brief

90 minutes, five steps

●	14:10	Welcome, decision problem, the data	15 min
●	14:25	Set up the project and read the data	5 min
●	14:30	Inspect: dictionary, structure, missingness	15 min
●	14:45	Prepare variables (age group, county)	7 min
●	14:52	Summarise one indicator	13 min
●	15:05	Visualise: a labelled graph	17 min
●	15:22	Record a limitation, share	11 min
●	15:33	Wrap-up, questions	7 min

Your workbook

R_Practical_1_inspect
_summarise_visualise.
Rmd has all the code,
explanations, self-
assessment questions
and FAQs.

We start with a question/decision, not always with the data

DECISION-MAKER

State COVID-19 response coordinator

DECISION

Where to prioritise extra hospital beds and staff in the next wave

POPULATION

People with a recorded COVID-19 diagnosis

PLACE

14 counties (our stand-in for districts)

PERIOD

First wave: 20 Jan to 30 Mar 2020

INTENDED ACTION

Flag counties where more diagnosed patients needed admission

Our indicator: percentage of patients diagnosed with COVID-19 who were admitted to hospital for COVID-19, by county of residence.

Synthea COVID-19 10K: synthetic patient records



Synthetic

Simulated from published disease models. No real person is in the data, so it can be shared freely for teaching.



Massachusetts, USA

14 counties and US hospitals. The numbers say nothing about African health systems; the methods transfer.



EHR-style linked tables

One row per event (visit, diagnosis, lab result), not one row per facility-month. DHIS2-style aggregates are built from data like these.

Today's file

`covid_patients.csv`

8,820

rows: one per patient diagnosed with COVID-19

8

columns (see the data dictionary)

0.8 MB

combined from 3 of the 16 files in the download

How the combined file was made

conditions.csv*one row per diagnosis*

Confirmed COVID-19
(code 840539006)

Denominator: 8,820 patients

patients.csv*one row per patient*

Birth and death date,
sex, county, ZIP

No names, ID numbers or addresses

encounters.csv*one row per visit*

Hospital admission for
isolation, for COVID-19

Numerator: 1,867 patients

**covid_patients.csv**

- Combined using the patient ID, which appears in all three files
- One row per diagnosed patient: no one counted twice
- admitted = TRUE if the patient had a COVID-19 admission
- Checked: still 8,820 rows; every patient matched a county

*Full code: workbook section 1.5 and make_covid_patients.R.
You will join files yourselves on Day 2.*

Six questions to ask before any analysis



Meaning

What does each variable mean? Read the data dictionary.



Structure and types

Dates read as dates? TRUE/FALSE as logical?

`glimpse()`, `head()`



Missing values

Where, how many, and does missing mean something?

`is.na()`



Size

How many rows and columns?

`dim()`



Unit of observation

What is one row? Is each ID unique?

`n_distinct()`



Ranges and categories

Impossible dates? Which categories are present?

`summarise()`, `count()`

Start with the data dictionary

Variable	Type	Meaning
<code>patient_id</code>	text	Unique patient identifier (synthetic)
<code>covid_date</code>	date	Date COVID-19 was diagnosed
<code>birth_date</code>	date	Date of birth
<code>death_date</code>	date	Date of death, any cause (blank if alive)
<code>sex</code>	text	M = male, F = female
<code>county</code>	text	County of the patient's address
<code>zip</code>	text	Postal (ZIP) code
<code>admitted</code>	TRUE/FALSE	TRUE if admitted to hospital for COVID-19

Why it matters

Names do not always tell you meaning. `death_date` is any cause of death, not only COVID-19. `admitted` is one specific admission type.

Full dictionary with sources: workbook section 1.4.

What is one row? Check before you count

```
dim(covid)
glimpse(covid)
head(covid)

covid %>%
  summarise(
    n_rows = n(),
    n_unique_ids = n_distinct(patient_id)
  )
```

8,820

rows

= 8,820 unique patient IDs: one row per patient

8

columns

3 dates, 3 text, 1 **TRUE/FALSE**, 1 ID

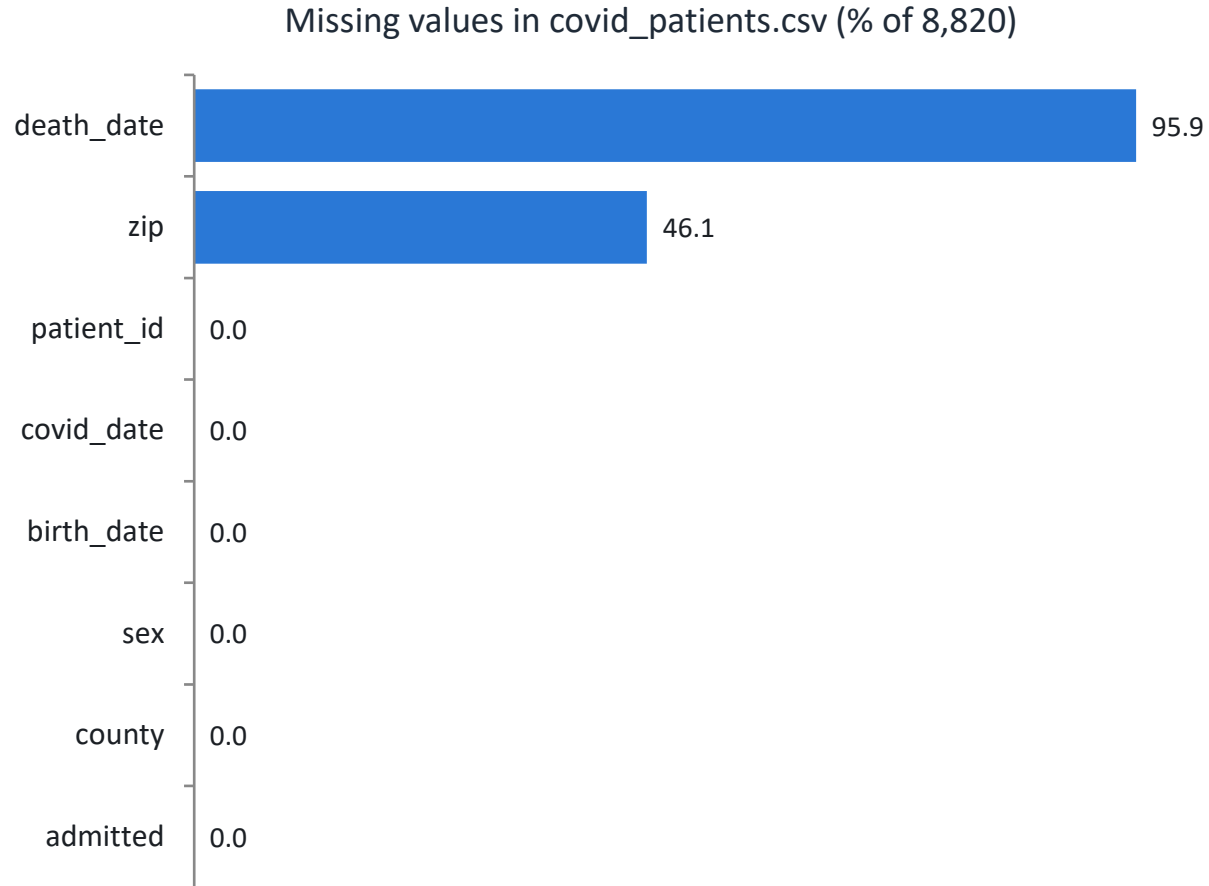
<date>

dates are real dates

so we can calculate age at diagnosis

Why it matters: if a patient appeared twice, they would be counted twice in both the numerator and the denominator.

Missing does not always mean a data problem



death_date 96% missing

Expected: blank means the patient was alive at the end of the simulation.

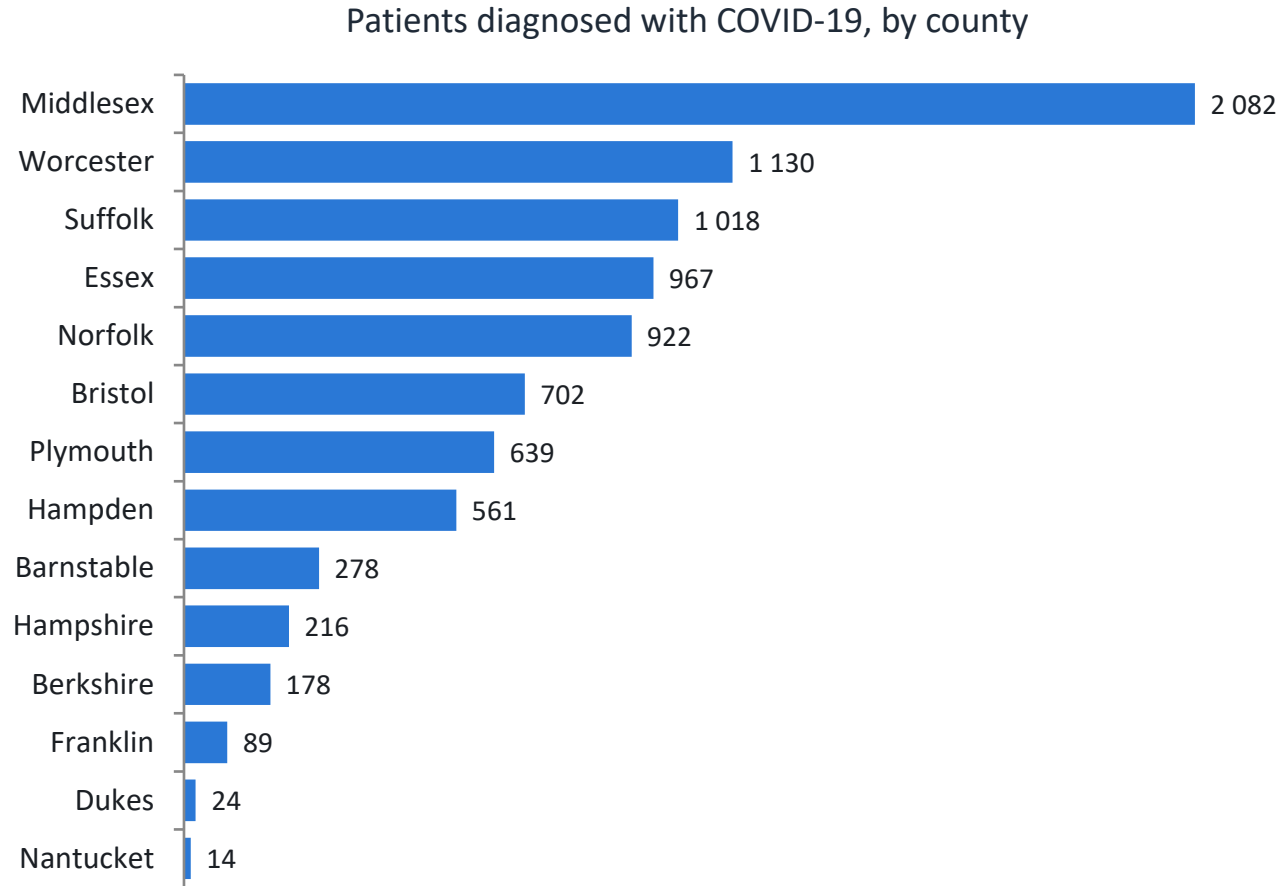
zip 46% missing

Would be a real quality problem if we needed it (for small-area maps). We use **county** instead.

county, sex, birth_date, admitted

Complete: the variables we need have no missing values.

Ranges and categories: spot the small counties



```
covid %>%  
  summarise(first = min(covid_date),  
            last  = max(covid_date))  
  
covid %>% count(county, sort = TRUE)  
covid %>% count(admitted)
```

- Period: 20 Jan to 30 Mar 2020
- 1,867 admitted, 6,953 not admitted
- Three counties have fewer than 100 patients: Franklin, Dukes, Nantucket

Derive variables that mean something for the question

```
covid <- covid %>%  
  mutate(  
    age = floor(time_length(  
      interval(birth_date, covid_date), "years")),  
    age_group = cut(age,  
      breaks = c(0,10,20,30,40,50,60,70,80,Inf),  
      labels = c("0-9","10-19", ... ,"80+"),  
      right = FALSE),  
    county = str_remove(county, " County$"),  
    sex = recode(sex, "F" = "Female", "M" = "Male")  
  )
```

Why age at diagnosis?

It is fixed at the time of the illness, so the code gives the same answer whenever it is run.

Age calculated from today's date would make everyone older every time you re-ran the analysis.

Age at diagnosis: median 41 years, range 0 to 110. Every patient has an age group.

Define the indicator before you calculate it

Numerator	Patients with ≥ 1 encounter coded Hospital admission for isolation (1505002) with reason COVID-19
Denominator	Patients with a recorded COVID-19 diagnosis (840539006)
Place	County of residence (14 counties)
Period	Diagnoses 20 January to 30 March 2020
Source	Synthea COVID-19 10K synthetic EHR data

```
overall <- covid_pts %>%  
  summarise(  
    diagnosed = n(),  
    admitted  = sum(admitted),  
    pct = 100 * admitted / diagnosed  
  )
```

TRUE counts as 1 and **FALSE** as 0, so **sum()** counts admissions and **mean()** gives the proportion.

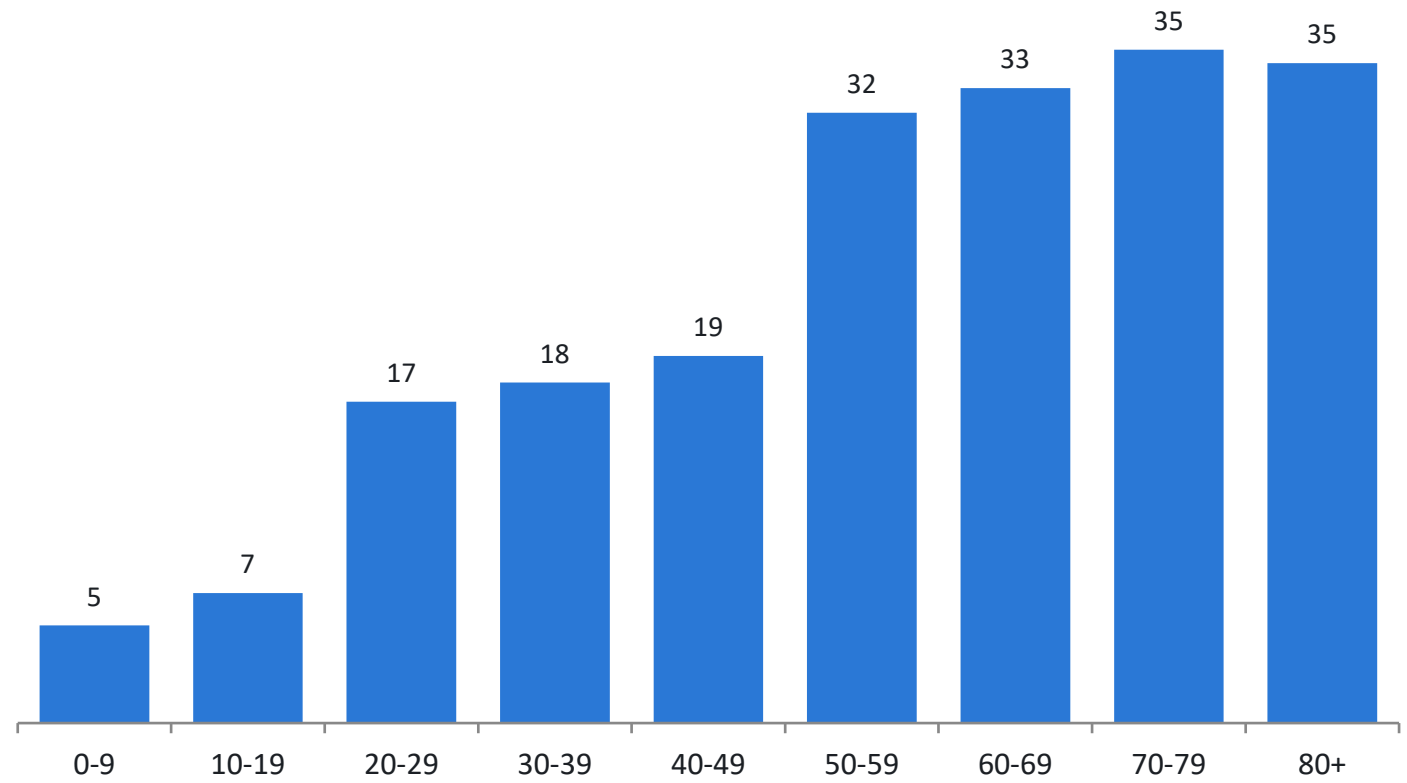
About 1 in 5 diagnosed patients was admitted

21.2%

of patients diagnosed with COVID-19
were admitted to hospital

1,867 admitted / 8,820 diagnosed

% admitted, by age group at diagnosis (years)



By county: `group_by()` then `summarise()`

```
by_county <- covid %>%  
  group_by(county) %>%  
  summarise(  
    diagnosed = n(),  
    admitted  = sum(admitted),  
    pct       = 100 * admitted / diagnosed,  
    .groups   = "drop"  
  ) %>%  
  arrange(desc(pct))
```

County	Diagnosed	Admitted	%
Franklin	89	22	24.7
Bristol	702	170	24.2
Barnstable	278	67	24.1
...			
Essex	967	180	18.6
Nantucket	14	2	14.3
Dukes	24	3	12.5

Checkpoint: look at the diagnosed column for the top and bottom counties. What do you notice?

Are the numbers big enough to trust?

If one more patient had been admitted, how much would the percentage change?

+7.1

percentage points

Nantucket

14 patients

+4.2

percentage points

Dukes

24 patients

+1.1

percentage points

Franklin

89 patients

+0.05

percentage points

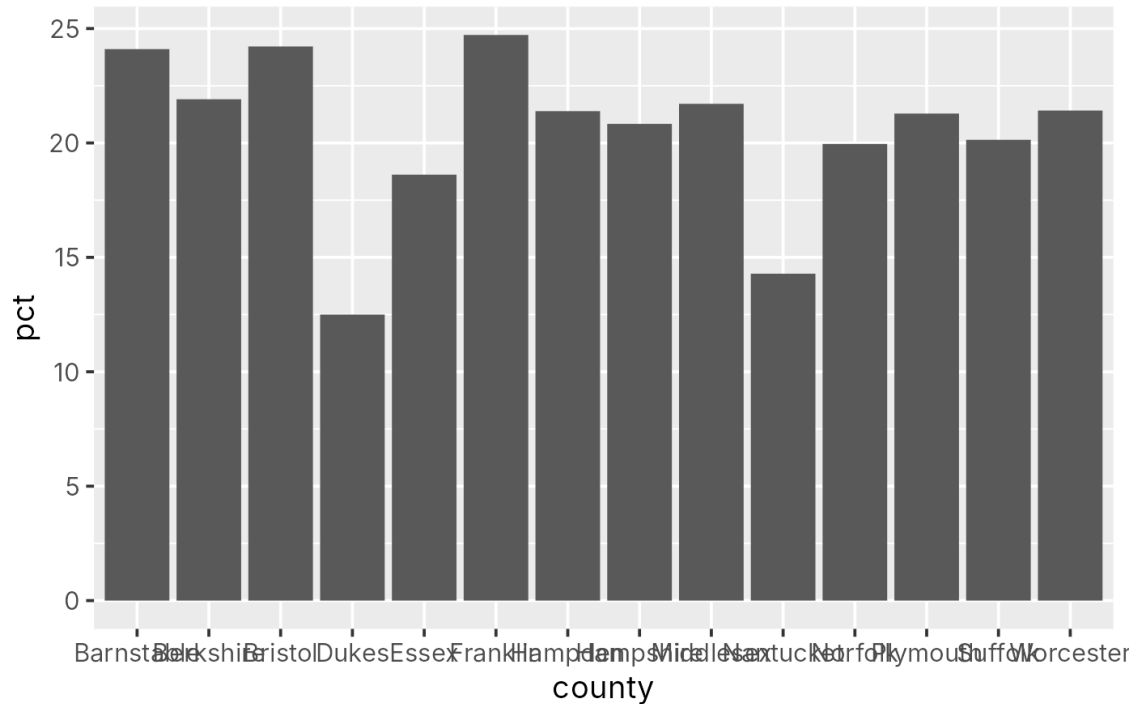
Middlesex

2,082 patients

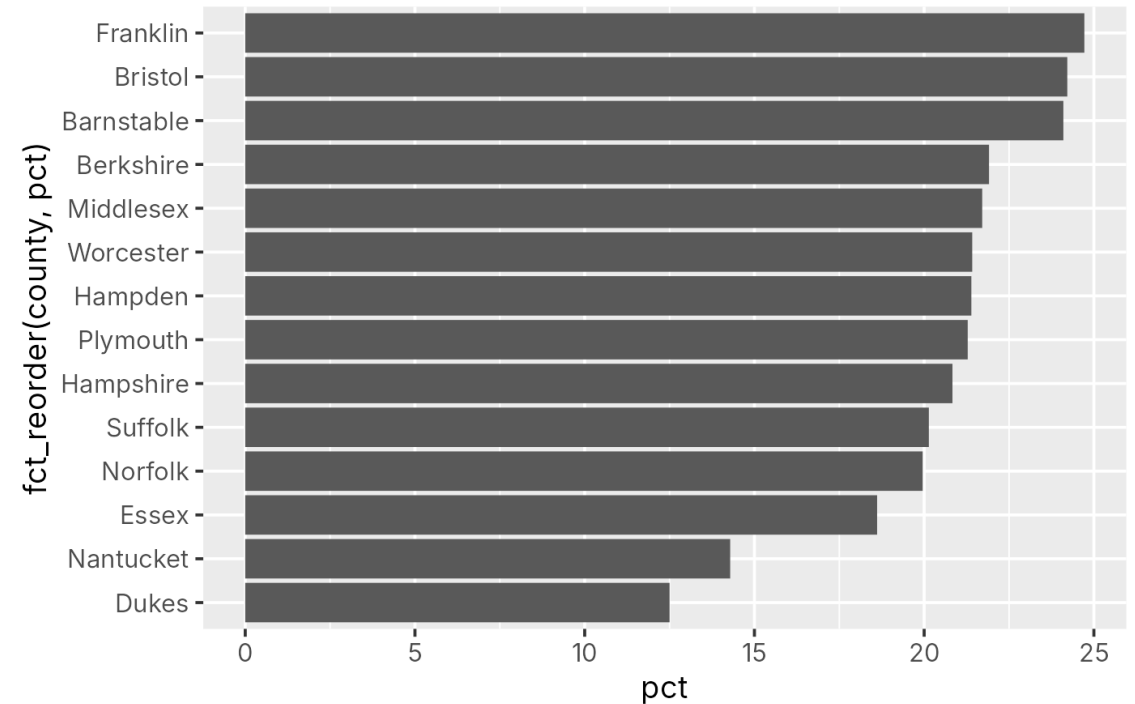
```
by_county <- by_county %>%  
  mutate(  
    one_patient = 100 / diagnosed, # % points per patient  
    small_n     = diagnosed < 100  # TRUE if under 100  
  )
```

A small county can look “best” or “worst” by chance. The cut-off of 100 is a rule of thumb for today, not a statistical law.

Build the graph one layer at a time

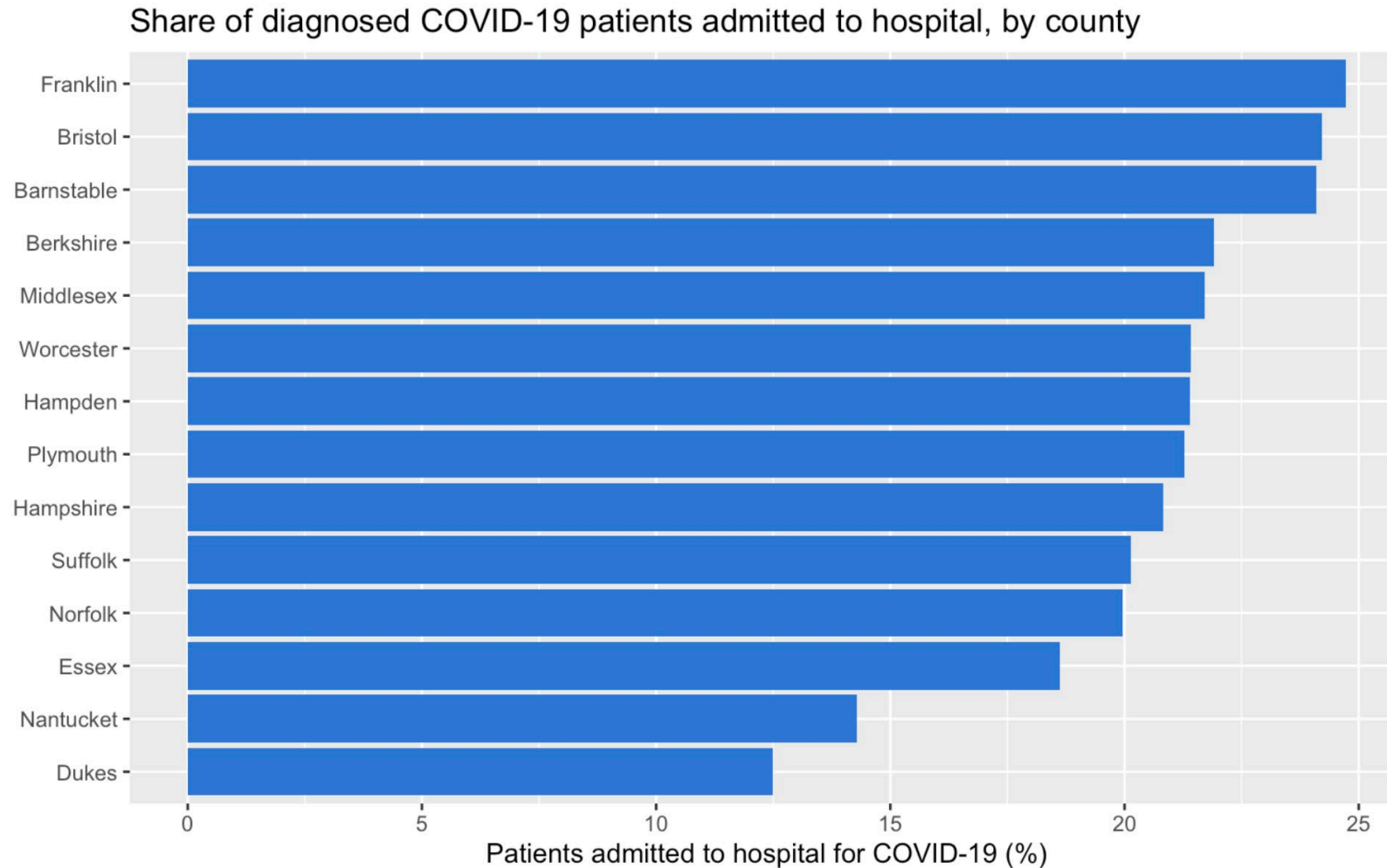


Layer 1: `ggplot(by_county, aes(county, pct)) + geom_col()` Names overlap, alphabetical order, no labels.



Layer 2: `aes(x = pct, y = fct_reorder(county, pct))` Horizontal and ordered by value: readable.

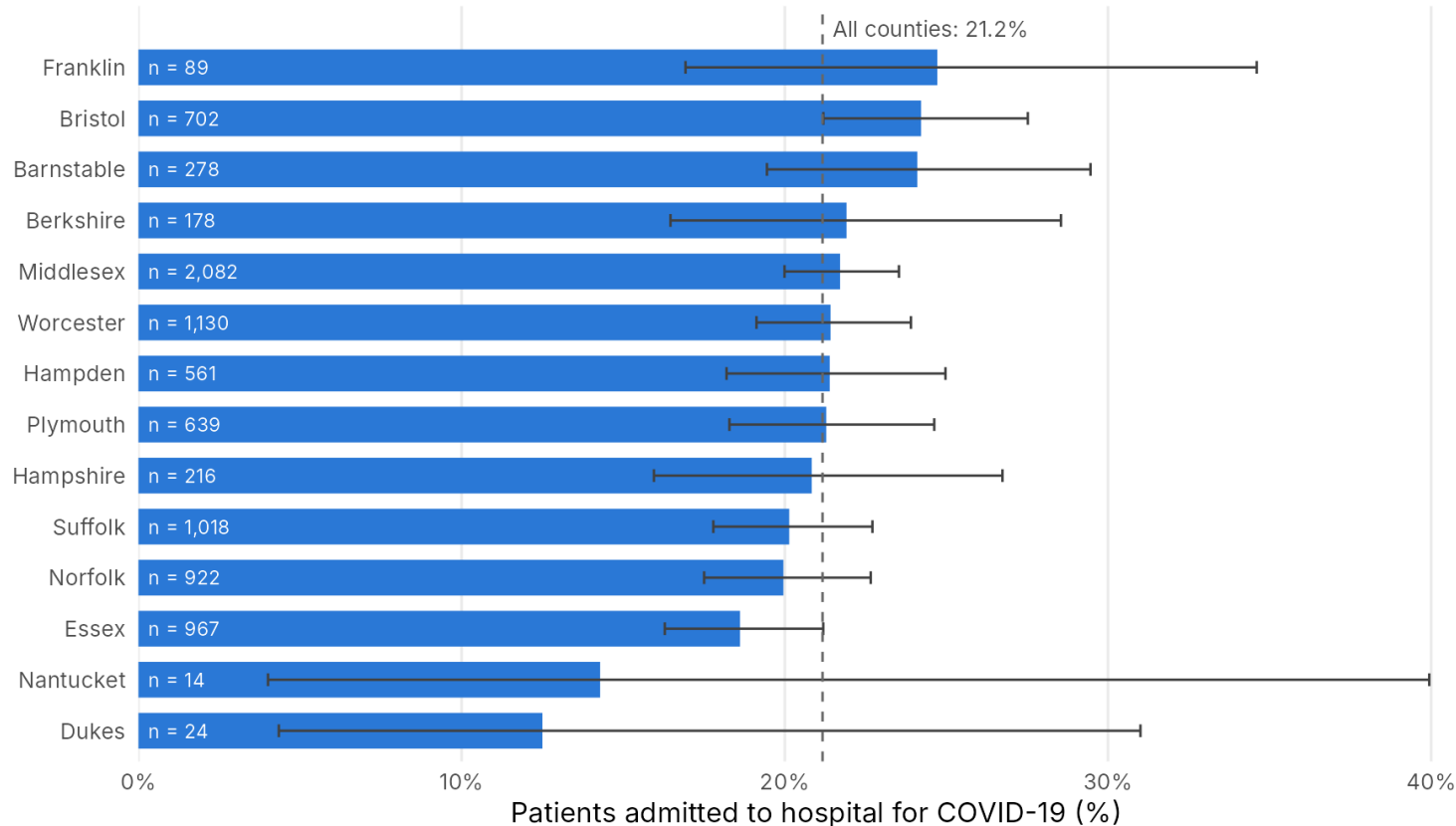
Build the graph one layer at a time



A graph someone outside the room can understand

Share of diagnosed COVID-19 patients admitted to hospital, by county

Bars show the percentage admitted; lines show 95% confidence intervals (Wilson).
n = number of patients diagnosed with COVID-19 in the county.



Source: Synthea COVID-19 10K synthetic dataset (Massachusetts, USA). Diagnoses 20 Jan to 30 Mar 2020.
Admission = encounter coded 'Hospital admission for isolation' (SNOMED 1505002) for COVID-19. Synthetic data: for training only.

Every label earns its place

- Title states the question
- Subtitle explains bars, lines and n
- Axis label with units (%)
- Denominator (n) on every bar
- 95% CIs: `geom_errorbar()`
- Reference line: state-wide value
- Caption: source, period, definition, “synthetic”
- One colour: colour adds no meaning here

A ranked chart always has a “worst” county

Most counties sit near 21%

The state-wide line runs through almost every interval.

Top and bottom = smallest n

Franklin (n = 89), Nantucket (n = 14) and Dukes (n = 24) have the widest intervals, all crossing the state-wide line.

Precise is not the same as unbiased

Large counties have narrow intervals, but age, testing access and coding still affect them.

Percentage is not volume

Bed need depends on the number admitted: Middlesex had 452 admissions, Franklin 22.

A useful limitation has three parts

What

is the limitation?

Why

does it matter for this indicator or decision?

What next

could be done about it?

Type	In today’s data
Representativeness	Synthetic Massachusetts population, not an African setting
Denominator	Only diagnosed people; testing access may differ by county
Comparability	Counties differ in age structure; no age adjustment
Precision	Nantucket (n = 14) and Dukes (n = 24): very wide intervals
Definition	Admission = one encounter code; real records are rarely this consistent

What you did today, and what comes next

Today

- Inspected three linked files: size, keys, missingness, codes
- Built a one-row-per-patient table with audited joins
- Summarised one indicator with 95% CIs
- Made and saved a fully labelled **ggplot2** graph
- Recorded a limitation alongside your outputs

After the session

Workbook section 8: eight self-assessment questions with hidden solutions.

Section 9: FAQs. Then click Knit to produce your report.

Next

16:10 BYOP Clinic 1: Decision Canvas for your own problem.

Day 2: data quality (2A) and reshape, join and audit (2B).

Questions?

Data: Synthea COVID-19 10K synthetic dataset, synthetichealth.github.io

Data dictionary: github.com/synthetichealth/synthea/wiki/CSV-File-Data-Dictionary

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